UID EX 6 **NAME: ADITHYA.B**

ROLL no :240701018

RAD (Rapid Application Development) is a software development methodology that emphasizes **quick development and iteration** of prototypes over strict planning and long development cycles. It's ideal for projects where requirements evolve through user feedback and where fast delivery is important.

Key Characteristics of RAD:

- Focus on **prototypes** instead of extensive documentation
- **User involvement** throughout the development process
- **Fast development cycles** with frequent iterations
- **Reusable components** to reduce effort
- Best suited for **UI/UX-intensive projects** like web and mobile apps

<u>Phase 1</u>: Requirements Planning

This is where the **project goals, system requirements, and user needs** are identified. Stakeholders, users, and designers collaborate to decide what needs to be built.

1) Identify Key Features

Navigation:

- Home
- Product Categories
- Product Listings
- Product Details
- Cart
- Order Confirmation

User Actions: Browsing, Searching, Adding to Cart, Checkout, Tracking Orders

2) Create a Requirements Document:

1. View list of product categories
2. View products under each category
3. View detailed information about a product
4. View and edit cart
5. Checkout and enter user/shipping/payment details
6. View order confirmation message

<u>Phase 2</u>: User Design

Designers and users collaborate to **create wireframes and prototypes**. The idea is to show how the application will look and behave before any actual coding starts.

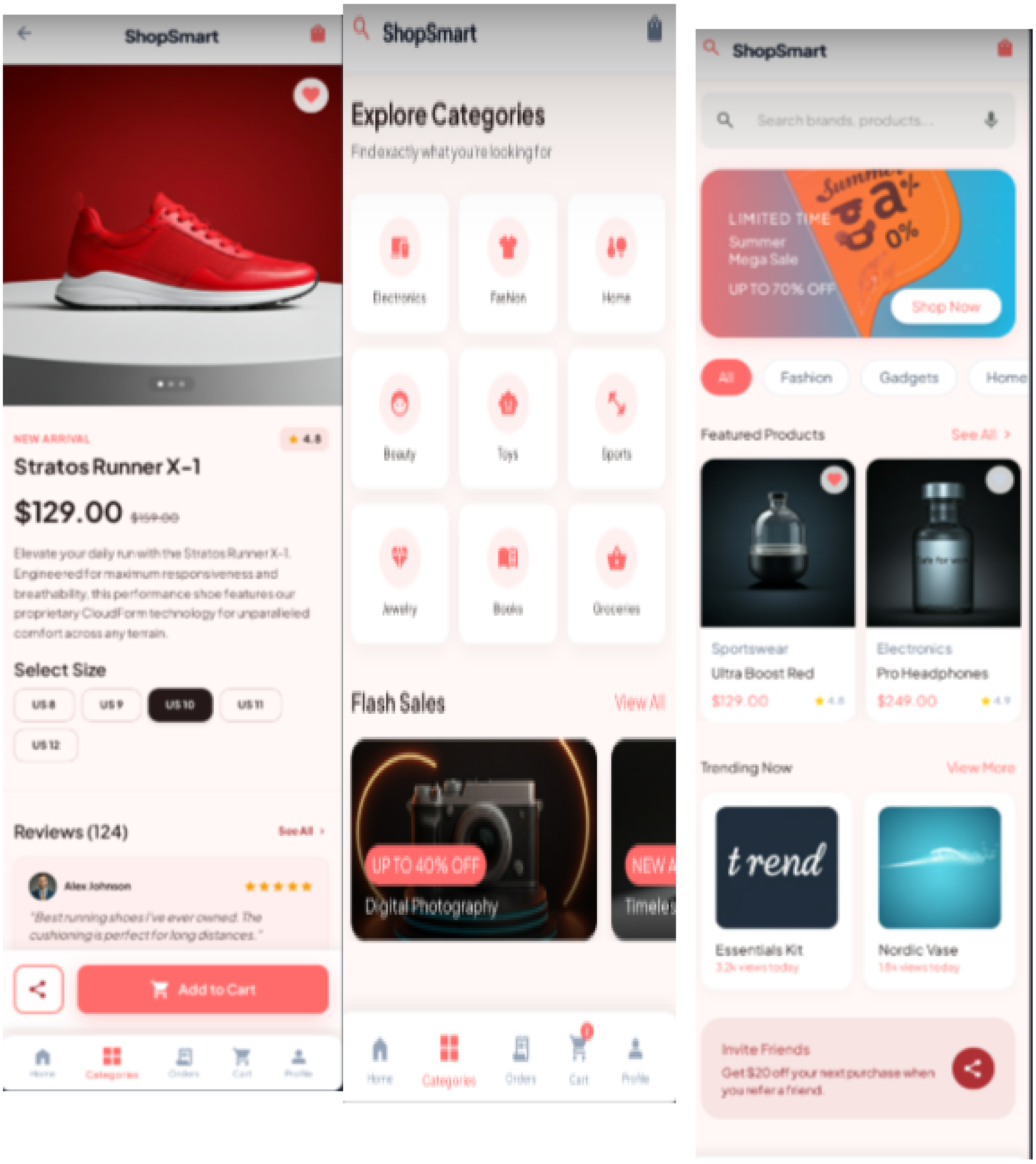
1) Tools Used: Axure: Used to design the wireframes and UI prototype.

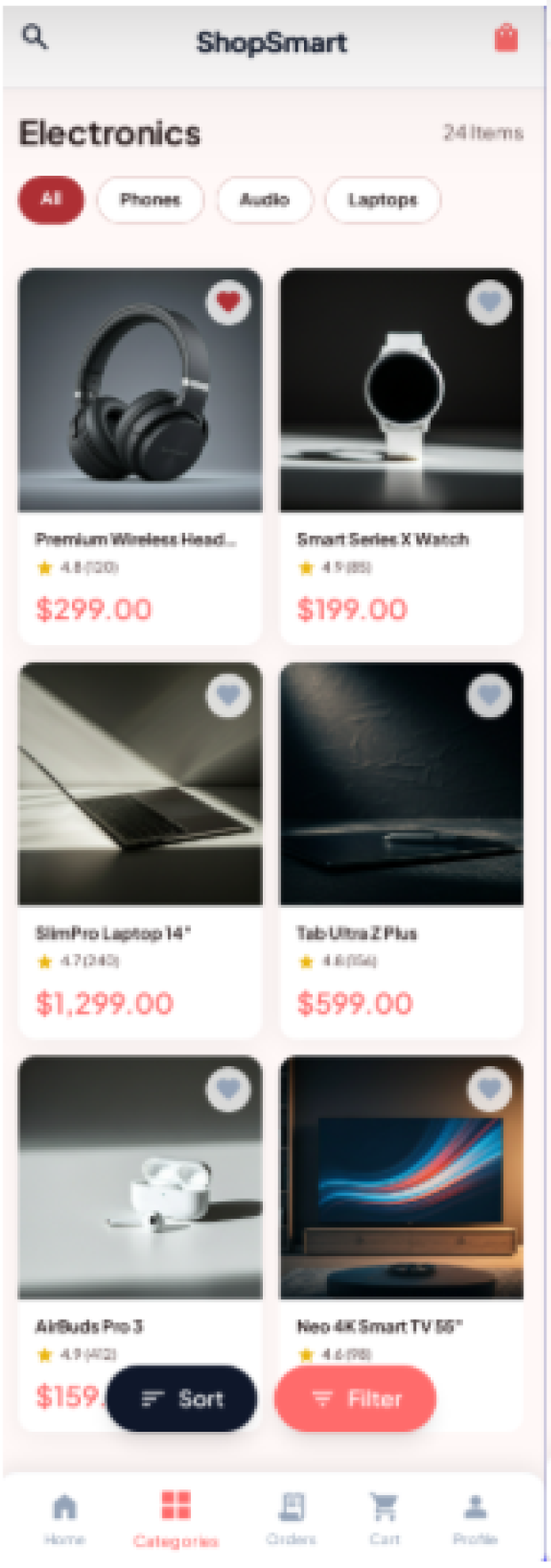
2) Wireframes Designed:

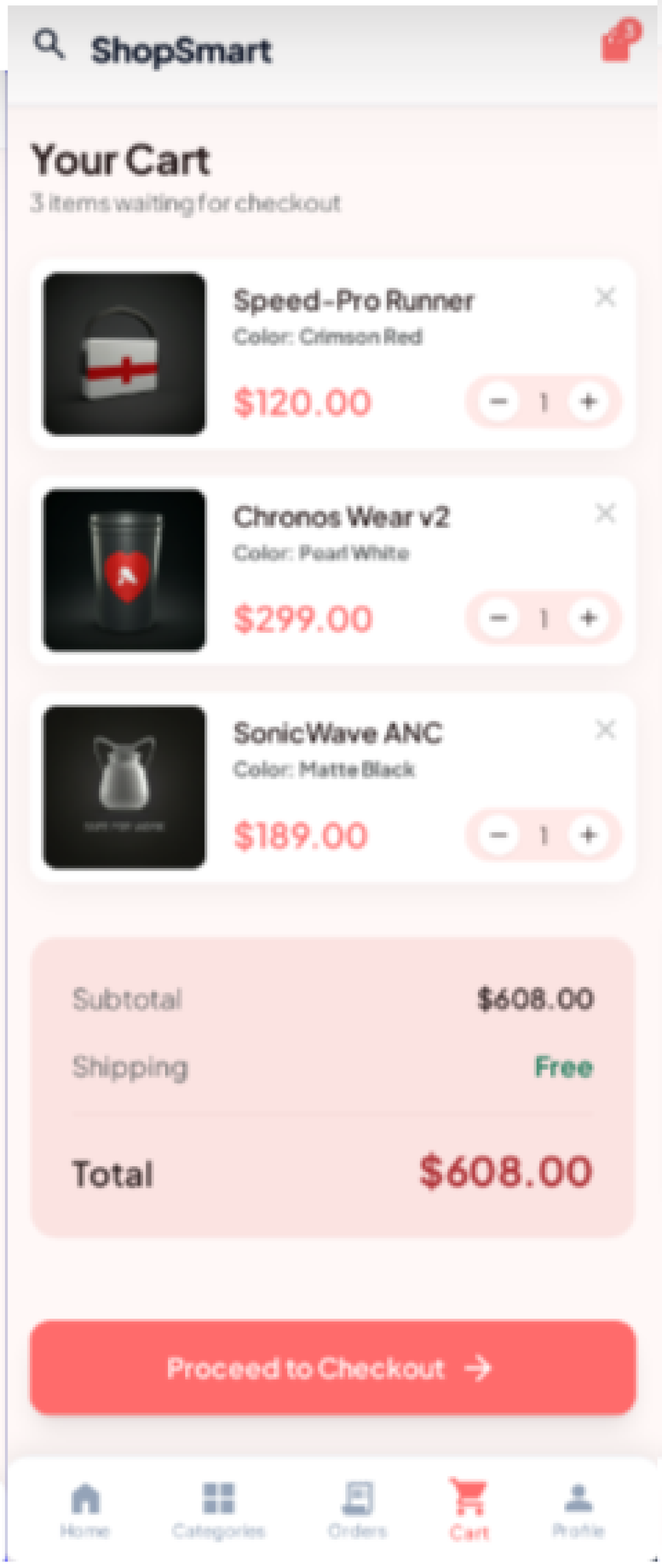
- **Home Page:** Highlights featured products and links to categories.
- **Product Categories:** Lists all product categories to explore.
- **Product Listings:** Shows products under a selected category.
- **Product Details:** Displays product info with "Add to Cart" option.

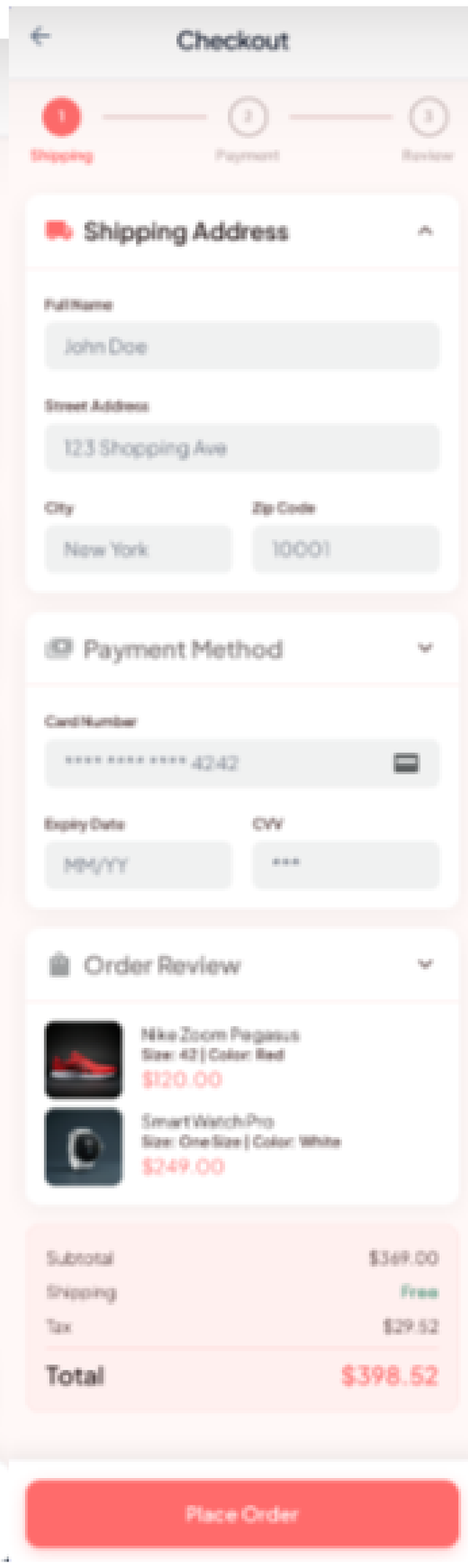
Cart: Shows added items with price and quantity details.
• **Checkout:** Collects shipping and payment information. •
Order Confirmation: Confirms successful order placement.

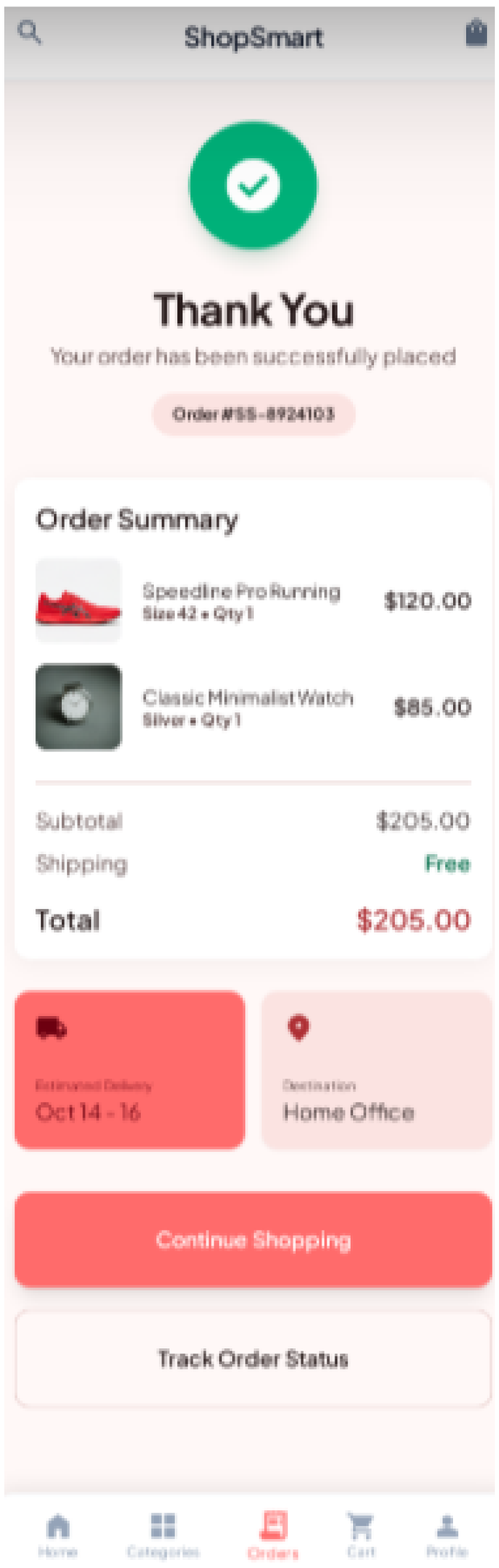
Designed **interactions and navigation** using Axure's Prototype mode
Reused components like headers/footers using **Components**
Annotated elements for clarity (e.g., what a button does, purpose of sections)




ShopSmart
Electronics
24 Items
All
Phones
Audio
Laptops
Premium Wireless Head...
4.8 (120)
$299.00
Smart Series X Watch
4.5 (85)
$199.00
SlimPro Laptop 14"
4.7 (240)
$1,299.00
Tab Ultra Z Plus
4.6 (154)
$599.00
AirBuds Pro 3
4.9 (412)
$159.
Sort
Neo 4K Smart TV 55"
4.6 (98)
Filter
Home
Categories
Orders
Cart
Profile


ShopSmart
Your Cart
3 items waiting for checkout
Speed-Pro Runner
Color: Crimson Red
$120.00
1
Chronos Wear v2
Color: Pearl White
$299.00
1
SonicWave ANC
Color: Matte Black
$189.00
1
Subtotal
$608.00
Shipping
Free
Total
$608.00
Proceed to Checkout
Home
Categories
Orders
Cart
Profile


Checkout
Shipping
Payment
Review
Shipping Address
Full Name
John Doe
Street Address
123 Shopping Ave
City
New York
Zip Code
10001
Payment Method
Card Number
**** **** **** 4242
Expiry Date
MM/YY
CVV
***
Order Review
Nike Zoom Pegasus
Size: 42 | Color: Red
$120.00
Smart Watch Pro
Size: One Size | Color: White
$249.00
Subtotal
$369.00
Shipping
Free
Tax
$29.52
Total
$398.52
Place Order


ShopSmart
Thank You
Your order has been successfully placed
Order #SS-8924103
Order Summary
Speedline Pro Running
Size 42 • Qty 1
$120.00
Classic Minimalist Watch
Silver • Qty 1
$85.00
Subtotal
$205.00
Shipping
Free
Total
$205.00
Estimated Delivery
Oct 14 - 16
Destination
Home Office
Continue Shopping
Track Order Status
Home
Categories
Orders
Cart
Profile

Phase 3: Construction

This is where the **actual development happens**, but in RAD, it is tightly coupled with ongoing testing and feedback. You refine the prototype based on user responses.

1) Interactive Prototypes:

- Created using Axure's **prototype linking** features.
- Functional paths designed:
  - From **Home → Product Categories → Product Listings → Product Details → Add to Cart → Checkout → Order Confirmation**

2) Dynamic Elements:

- Simulated transitions and modals for adding to cart and checking out. • Screens simulate real-time behavior such as navigating between product listings and cart.

3) Testing and Iteration:

- Prototype previewed with **"Present"** mode in Figma.
- Collected feedback from stakeholders (assumed for documentation). • Iterated based on visual hierarchy, accessibility, and flow suggestions.

Phase 4: Cutover

In this final phase, the system goes live. This includes **finalizing the design**, **training users**, and **preparing documentation** for developers.

1) Finalization and Export:

- Final design prototype shared through Figma link.
- Ready for hand-off to development team.
- Could be exported to **HTML/CSS** using plugins like Figma-to-HTML or further tools if needed.

2) User Training & Support:

- Walkthrough of the prototype for stakeholders or development team.
- Notes/documentation provided for developers to understand functionality and user flow.
- Easy access to Figma comments for further collaboration and questions.